

Hyperloop Development Program

Reporting Period 6: Jul 2023 – Dec 2023

13-05-2024

V1.0

This report highlights the progress of the Hyperloop Development Program. First section summarizes the highlights within the growing hyperloop market, the second section provides an overall program overview, followed by section of progress per working group, and summarized in the financial overview.

The Hyperloop Development Program has requested a renewal extension, showing progress despite funding delays and permitting challenges, with the construction of the EHC underway and new partnerships established. The plan has been adjusted, incorporating changes in Working Groups deliverables, budgets, and time estimations.

REPORTING ON HIGHLIGHTS WITHIN HYPERLOOP MARKET

The past months have shown highlights in the Global developments in the Hyperloop Market:

- Jul:
 - o Hardt Hyperloop announces closing of €12M investment round. The investment was made by a consortium of public & private investors.
 - o The European Hyperloop Week (largest student hyperloop competition), was held in Edinburgh
 - o Hyperloop article released in the European Parliament magazine, The Parliament
- Aug:
 - o Hardt Hyperloop to release its technical progress paper on the 10 year anniversary of hyperloop of Elon Musk's hyperloop concept paper
 - o Hyperconnected Europe initiative is growing with Antwerp, the City region of Turnhout, and the region Stedendriehoek.
 - o Foundations for the European Hyperloop Center have been completed
- Sep:
 - o All linear pipes for the European Hyperloop Center have been completed by Mercon
 - o The European Hyperloop Center had its kick-off event with the installation of the first two pipes
 - o The Hyperloop Conference was held in Gdansk, Poland.
- Oct:
 - o Hardt Hyperloop and Zeleros hyperloop sign MoU for a strategic collaboration
 - o The Business Booster was held by InnoEnergy, featuring the three hyperloop startups Zeleros, Nevomo, and Hardt Hyperloop. Visit by vice-president of the European Commission, Maros Sevcovic.
 - o The lane-switch pipes have been completed by Mercon
- Nov:
 - o All the pipes for the European Hyperloop Center have been placed by Denys
 - o Spatial Integration Report of hyperloop has been handed over by the Hyperloop Development Program to the Dutch Minister of Infrastructure and Water Management, Mark Harbers
- Dec:

- US Hyperloop company Hyperloop One ceases operations after new investments were not made in the company
- Dutch Province Limburg announces ambition to build hyperloop between Heerlen and Aken
- Hardt Hyperloop signs contract with Indian hyperloop company TuTr for providing magnet propulsion drives

REPORTING ON OVERALL PROGRAM PROGRESS

This report is to confirm costs incurred and progress in the Hyperloop Development Program over the period Jul 2023 – Dec 2023.

The key program deliverable and key deliverable for this stage per working group are represented below.

WG	Key Program deliverable	Key deliverable Status
WG Cargo	Demonstrated hyperloop for cargo system (TRL-7) in the EHC	Vehicle integration completed, currently in commissioning
WG EHC	European Hyperloop Center (EHC) Phase A completed	Guideway construction completed, current installing auxiliaries.
WG FP	Identify and develop 5 routes up to a preliminary level, of which 1 with a go ahead build-out of route	All Preliminary route studies completed. But not yet a commitment for a pilot.
WG Passengers	Passenger hyperloop system concept (TRL-3 & SRL-5)	System Concept Design Definition (on-hold for now)
WG Program Management	Establish a connected hyperloop ecosystem	Extension requested, with new deliverables for 2024.

HDP is continuing Stage 3 of a total of 4 stages – as there have been encountered delays, the stage 3 is extended till Jul 2024. The progress of the Working Groups and Work Packages is presented in figure 1. Green means the Work Package is delivered; yellow the Work Package is on-going.

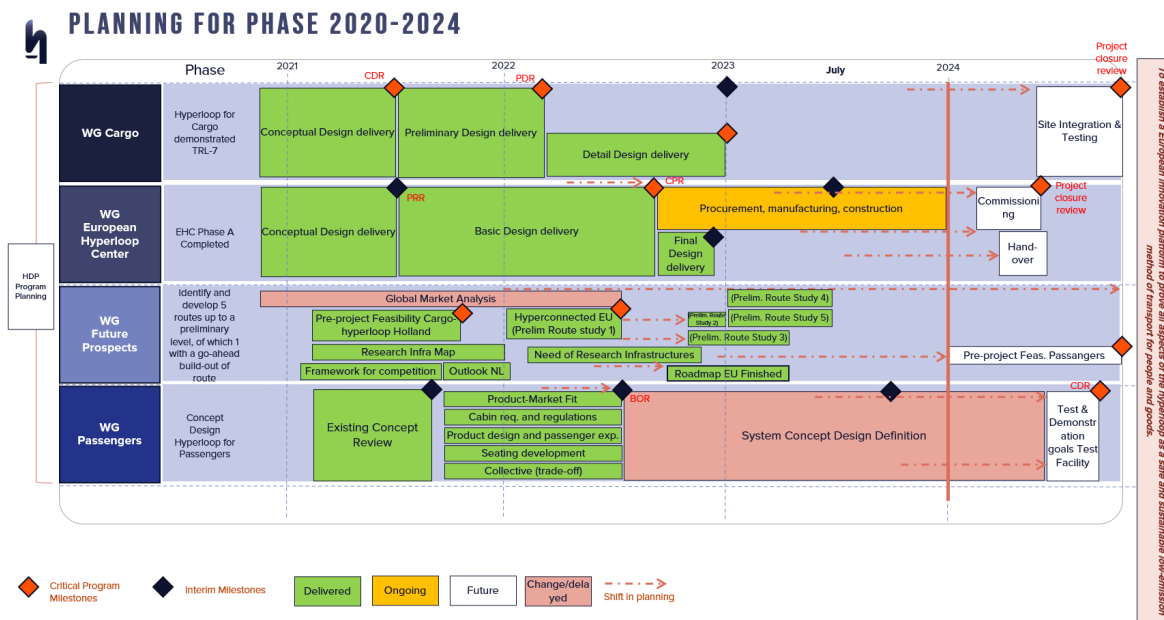


Figure 1: Stages & Milestone planning HDP

The detailed overview of program deliverable status for open stage 1, stage 2, and stage 3 is presented below. A number of milestones from stage 1 and 2 have encountered delays, all adjectives delivery dates are presented below.

#	Milestone	Working Group	Status	(Adj.) Date delivery
Open from Stage 1:				
1.9	Final Market opportunity attractiveness	Future Prospects	Ongoing	Q4-2024
Open from Stage 2:				
2.4	Partners/suppliers contracted for Manufacturing & Assembly.	EHC	Completed	Q4 2023
2.8	Scheme of operations for Hyperloop Research Infrastructure	Future Prospects	On-going	Q2 2024
Stage 3:				
3.5	Preliminary Route #2 (Hyperconnected Europe)	Future Prospects	Finished	Q4 2022
3.6	Preliminary Route #3 (Delhi – Ahmedabad)	Future Prospects	Finished	Q1 2023
3.7	Preliminary Route #4 (Italy)	Future Prospects	Finished	Q2 2023
3.8	Preliminary Route #5 (The Netherlands)	Future Prospects	Finished	Q2 2023
3.9	EHC Delivery & Start first tests	EHC	Not Started (Delayed)	Q2 2024
3.10	Project hand-over EHC	EHC	Not Started (Delayed)	Q2 2024
3.11	Roadmap EU Finished	Future Prospects	Finished	Q1 2023
3.12	Work Package Research Infra finished	Future Prospects	Not Started (Delayed)	Q4 2023

REPORTING PER WORKING GROUP

WG Cargo

Status: Proceeding according to WG Plan

Within the WG Cargo major progress has been made with respect to the manufacturing, assembly and delivery of the European Hyperloop Center. Highlights include:

- Major infrastructure components were received such as switch floor, door, expansion joints, and gaskets.
- Switch guideway production & assembly was finished by Mercon.
- Denys aligned all guideway interfaces on-site (sliders & x-stoppers).
- Finished all vehicle design scopes (exl. fairings).
- An inductive gap sensor system was developed and tested.
- All magnets (propulsion, levitation & guidance) have been delivered.
- All magnet drives for propulsion, levitation and guidance were assembled.
- Batteries for the vehicles were delivered.
- Delivery of bogie frames & start of vehicle assembly.
- Vehicle assembly completed by Hardt.
- Electrical infrastructure has been prepared.

Below is a summary in images:



Figure 1: Production of switch pipes.



Figure 2: Assembled Switch Pipe



Figure 4: Vehicle assembly.

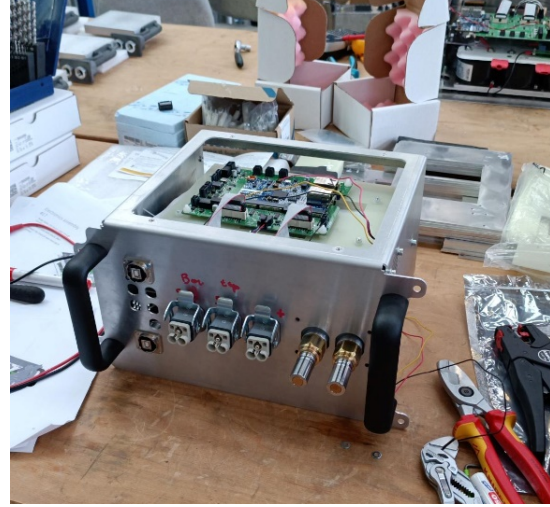


Figure 3: Completed magnet drive.

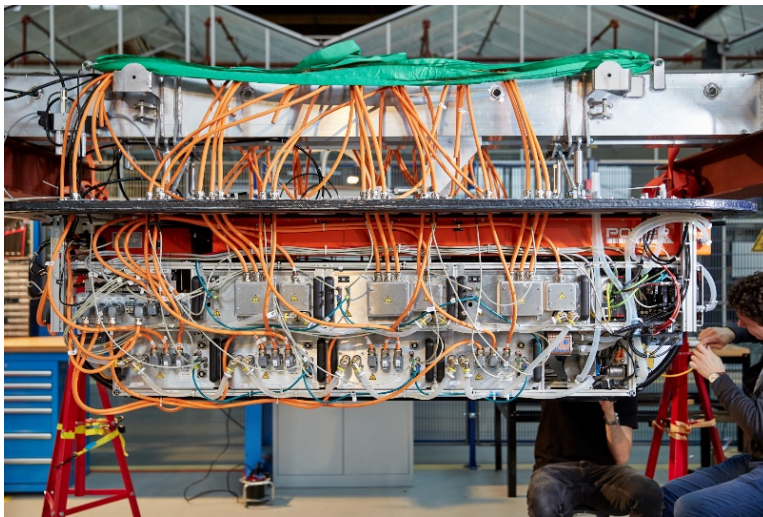


Figure 5: Completed electronics bay on vehicle.



Figure 6: Vehicle assembly.

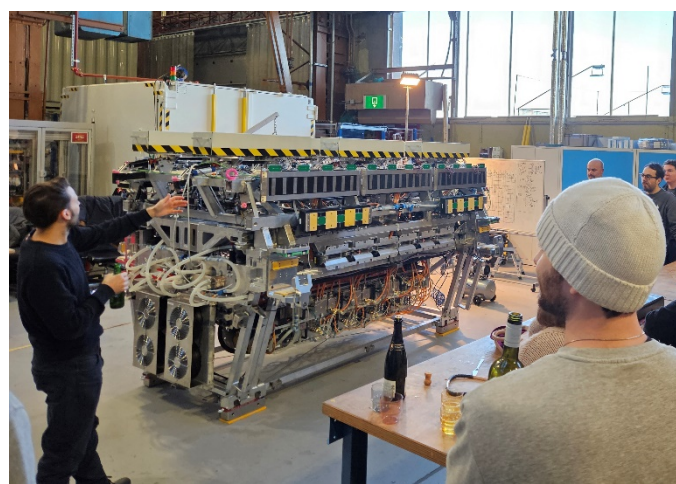


Figure 7: Completely assembled vehicle.

WG Passengers

Status: On-hold

Almost no progress was made towards full-scale hyperloop passenger design due to the focus on the European Hyperloop Center and WG Cargo.

Progress was made towards standardization and regulation of the passenger hyperloop. During H2-2023 progress was made regarding hyperloop vocabulary definitions, which were submitted for voting and accepted by the JTC20.

New work items and a new work group was proposed within the JTC20 regarding hyperloop safety. This proposal was prepared by Hardt, Zeleros and Transpod and was well received by the other members of the JTC20. It is expected that work regarding this topic will commence later in 2024.

WG EHC

Status: On-track to complete the test-facility by 31st of March 2024 handover planned in Q2

For the EHC in Groningen - During H2 2023 all pipes (linear & switch) were delivered by Mercon to Veendam and on-site guideway installation was completed by Denys. Vacuum pumps were delivered on-site. Completed procurement of the control room, workshop and electrical.

In terms of permitting we came to an agreement with Veendam that no permit is further required. In September 2023, an official kick-off event occurred to celebrate the installation of one of the first hyperloop pipes. This marked a significant milestone and represented the initial phase towards constructing and completing the entire 420 meters.

Planned for H1 2024 is the completion of the test-facility to start testing with Hardt's vehicle.

For the Demotube Test-Facility in Switzerland – the subsystem development (Airlock, Vacuum assurance system, Foundation) - in the first phase of the DemoTube project, key components were developed, including the manufacturing of the airlock and the valve, essential for the system's operation. Additionally, this stage involved extensive preparations for construction, laying the groundwork for the project's infrastructure. The launcher system has also been developed during this phase, preparing the system and its manufacturing in 2024. This foundational work encompassed project management, securing partnerships, and preparing the test site, crucial steps for the project's progression.



Figure 10: Guideway interfaces.



Figure 9: Installed guideway.



Figure 8: Installation of guideway.

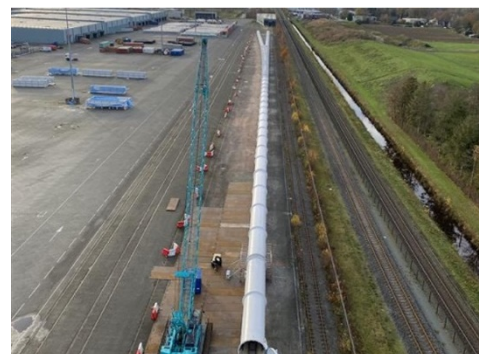


Figure 11: Installed guideway.



Figure 14: EHC kick-off event



Figure 15: EHC kick-off event



Figure 13: EHC kick-off event



Figure 12: EHC kick-off event

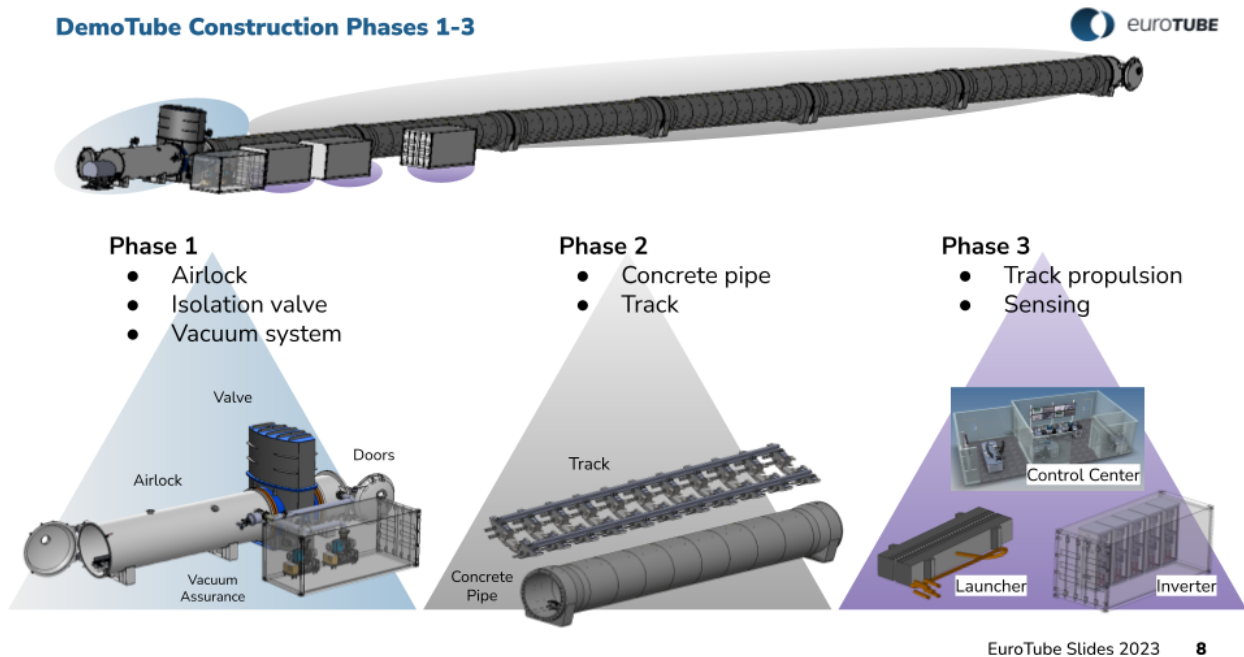


Figure 16: DemoTube Contrustion Phases 1-3

WG Future Prospects

Status: Ongoing

The following activities are summarized:

- **Expansion of Lead Generation and Opportunity Phases**
In this period 5 new potential routes have been added with local stakeholder engagement:
 1. Limburg – Aachen (Netherlands and Germany)
 2. Zeeland (Netherlands)
 3. Flanders (Belgium)
 4. Regio Stedendriehoek (Netherlands)
 5. Vienna (Austria)
- **Workshops on Hyperloop's Impact on Spatial Planning**
Following up on the workshops organized with the The Ministry of Infrastructure and Water Management (IenW), the impact of hyperloop technology on the spatial planning of the Netherlands for 2050 has been further explored. A report has been launched at the Dutch Innovation Expo 2023.
- **Collaboration with Dutch Railways and ProRail**
HDP engaged in collaborations with the Dutch Railways (NS) and ProRail to assess the potential impact of hyperloop technology on the Dutch rail network. This included conducting internal workshops and integrating hyperloop into ProRail's transport modeling platforms for independent route development studies. The results were showcased at the CVS conference in Brussels in November 2023.
- **Development of the Spanish Network**
A collaboration City of Zaragoza and the Region of Aragon has resulted in a letter of support for

developing the European hyperloop network. Discussions are pending on a commercial pilot and a study for a long-distance network, which would include Zaragoza's airport and connections to Pau and Toulouse.

- **Building International Relationships for Ecosystem Expansion**

HDP has actively worked on establishing relationships with potential partners to broaden the hyperloop ecosystem and assess potential routes on an international scale, including opportunities in India, Saudi Arabia, Dubai, and Türkiye. Discussions involve establishing twins of the European Hyperloop Center and ensuring we can build on each other's R&D base. This is also preparatory work for the next phase of the Hyperloop Development Program where R&D is performed and integrated in a global program.

- **Signing Letter of Intent with EuroTube**

HDP, EHC, and the Swiss EuroTube Foundation have signed a Letter of Intent to create a European hyperloop research infrastructure cluster to foster collaboration between EuroTube's DemoTube, located in Switzerland, and the European Hyperloop Center, based in the Netherlands.

- **Potential analysis for Hyperloop in Switzerland**

For the Federal Office of Transport (FOT) in Switzerland, EuroTube conducted a potential analysis for vacuum transport in Switzerland. This included a life-cycle analysis with focus on energy consumption and environmental impact of a vacuum transport infrastructure. This led to the creation of tools for LCA, energy and cost estimates that can be applied to other route studies.

- **Align European Test facilities into a collaborative cluster**

Alignment in testing facilities within Europe as one cluster. Based on the European space agency model the goal is to enhance, multinational collaboration, diverse expertise for example between the concrete and steel type structure of both current testing facilities.

The goal of this international cooperation is to enable joint projects and breakthrough missions in the future.